

INSURANCE ANALYTICS · SQL + PYTHON PROJECT

Customer Lifetime Value Insights & Recommendations

Customer Segmentation, CLV, Claims & Marketing Response Analysis

9,134

Customers

\$8,005

Avg. CLV

14.32%

Response Rate

\$93.22

Avg. Monthly Premium

5

States

Data Source: IBM Watson Marketing Customer Value Analysis Dataset (9,134 records)

Tools Used: PostgreSQL (aggregation, CTEs, subqueries, window functions), Python

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01 Executive Summary

This report presents a comprehensive analysis of the IBM Watson Marketing Customer Value dataset (9,134 auto insurance policyholders across 5 states). Using PostgreSQL — aggregate functions, GROUP BY/HAVING, CASE WHEN logic, subqueries, Common Table Expressions (CTEs), and window functions (RANK, DENSE_RANK, NTILE, LAG/LEAD) — the analysis investigates customer segmentation, Customer Lifetime Value (CLV), premium revenue, claims behavior, marketing response, and profitability risk.

The portfolio generates **\$73.1M** in total customer lifetime value and **\$851K** in monthly premium revenue, against **\$3.96M** in total claims. The average customer carries a CLV of **\$8,005** but pays only **\$93.22/month** in premium — and the data reveals a large gap between the highest- and lowest-value segments. Special Auto and Personal Auto policyholders carry the highest CLV; Luxury vehicle owners file the largest claims; and the Agent sales channel converts marketing responses at nearly double the rate of every other channel. Overall campaign response sits at a modest **14.32%**, with a small but valuable pool of high-CLV customers who have not yet responded — a clear opportunity, detailed in Section 9.

Snapshot at a Glance

9,134 Total Customers	5 States	3 Policy Types	\$8,005 Avg. CLV
\$851K Premium Revenue	\$3.96M Total Claims	14.32% Response Rate	\$37,657 Avg. Income

02 Dataset Overview & Key Metrics

The dataset (WA_Fn-UseC_-Marketing-Customer-Value-Analysis.csv) was loaded into a PostgreSQL table (customer_insurance) with 24 attributes covering customer demographics, policy details, claims history, and marketing response. Data quality checks in the SQL script confirmed **zero duplicate customer IDs and zero NULL values** across all columns — this is a clean dataset, and the cleaning script's text-standardization and type-casting steps served as defensive checks rather than corrections to missing data.

Metric	Value
Total Customers	9,134
States Covered	5 (California, Oregon, Arizona, Nevada, Washington)
Policy Types	3 (Personal, Corporate, Special Auto)
Coverage Levels	3 (Basic, Extended, Premium)
Total Customer Lifetime Value	\$73,117,126.30
Average Customer Lifetime Value	\$8,004.94
Total Monthly Premium Revenue	\$851,465
Average Monthly Premium	\$93.22
Total Claim Amount	\$3,964,967.05
Average Claim Amount	\$434.09
Average Customer Income	\$37,657.38
Overall Campaign Response Rate	14.32% (1,308 of 9,134)

Sample SQL — Response Rate Calculation (CASE WHEN via FILTER + Aggregation):

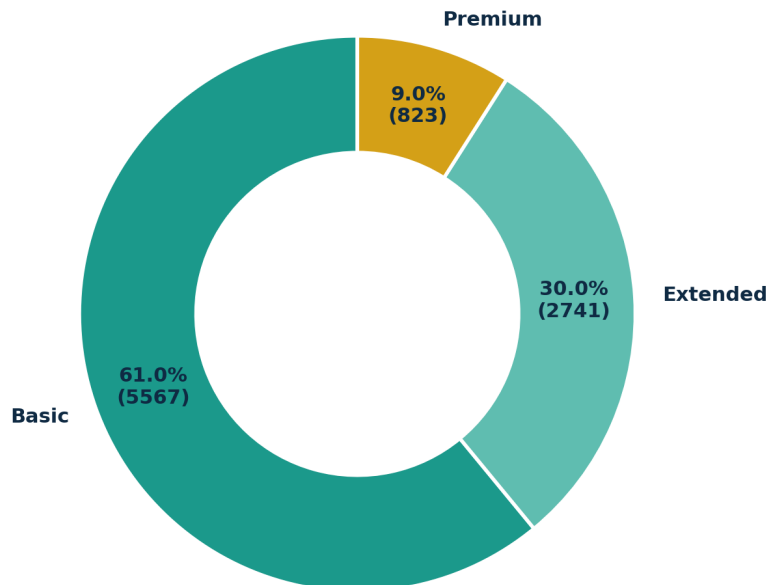
```
SELECT COUNT(*) AS total_customer,
       COUNT(*) FILTER (WHERE response = 'Yes') AS responders,
       ROUND(COUNT(*) FILTER (WHERE response = 'Yes') * 100.0 / COUNT(*), 2) AS response_rate
FROM customer_insurance;
```

03 Customer Segmentation

Segment-level GROUP BY queries profiled the customer base across state, policy type, coverage, employment status, and vehicle characteristics.

Segment	Top Category	Share
State	California	3,150 customers (34.5%)
Policy Type	Personal Auto	6,788 customers (74.3%)
Coverage	Basic	5,567 customers (61.0%)
Employment Status	Employed	5,698 customers (62.4%)
Vehicle Class	Four-Door Car	4,621 customers (50.6%)
Sales Channel	Agent	3,477 customers (38.1%)
Marital Status	Married	5,298 customers (58.0%)
Location Code	Suburban	5,779 customers (63.3%)

Customers by Coverage Level

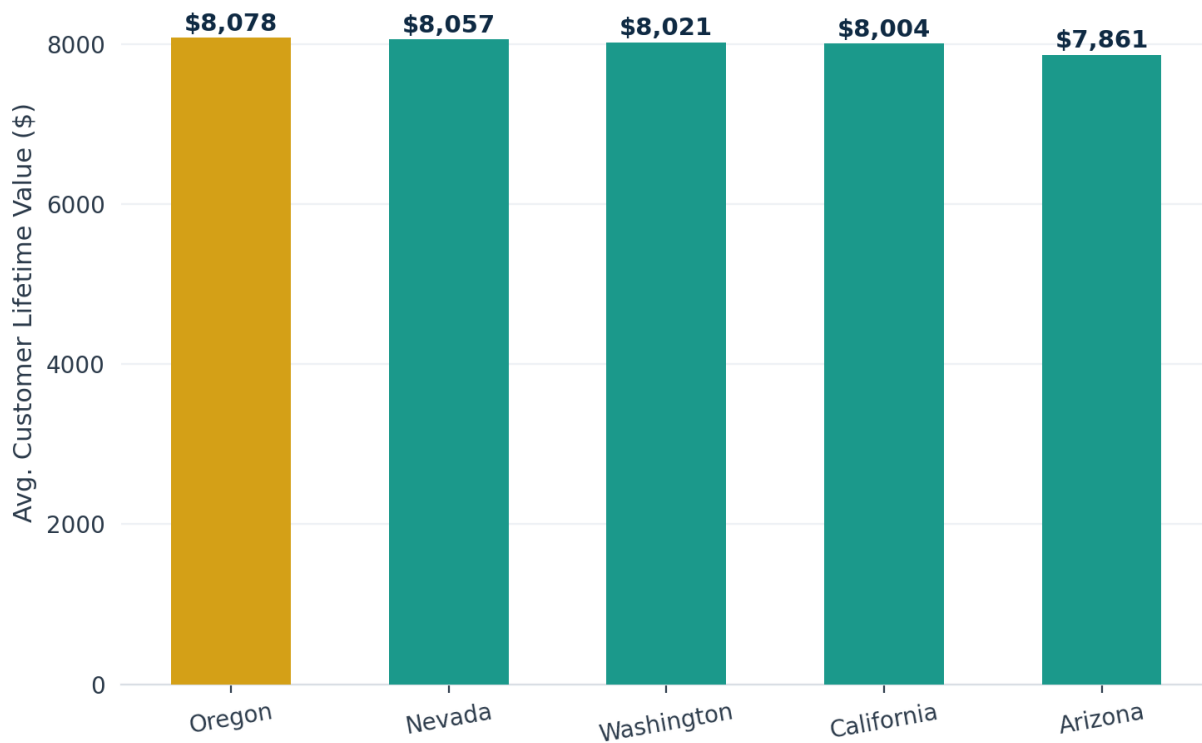


Insight: The book of business is heavily concentrated in Basic coverage (61%) and Personal Auto policies (74%) — a mass-market, price-sensitive customer base rather than a premium-skewed one. Only 9% of customers hold Premium coverage, representing the smallest but (as later sections show) most profitable segment.

04 Customer Lifetime Value Analysis

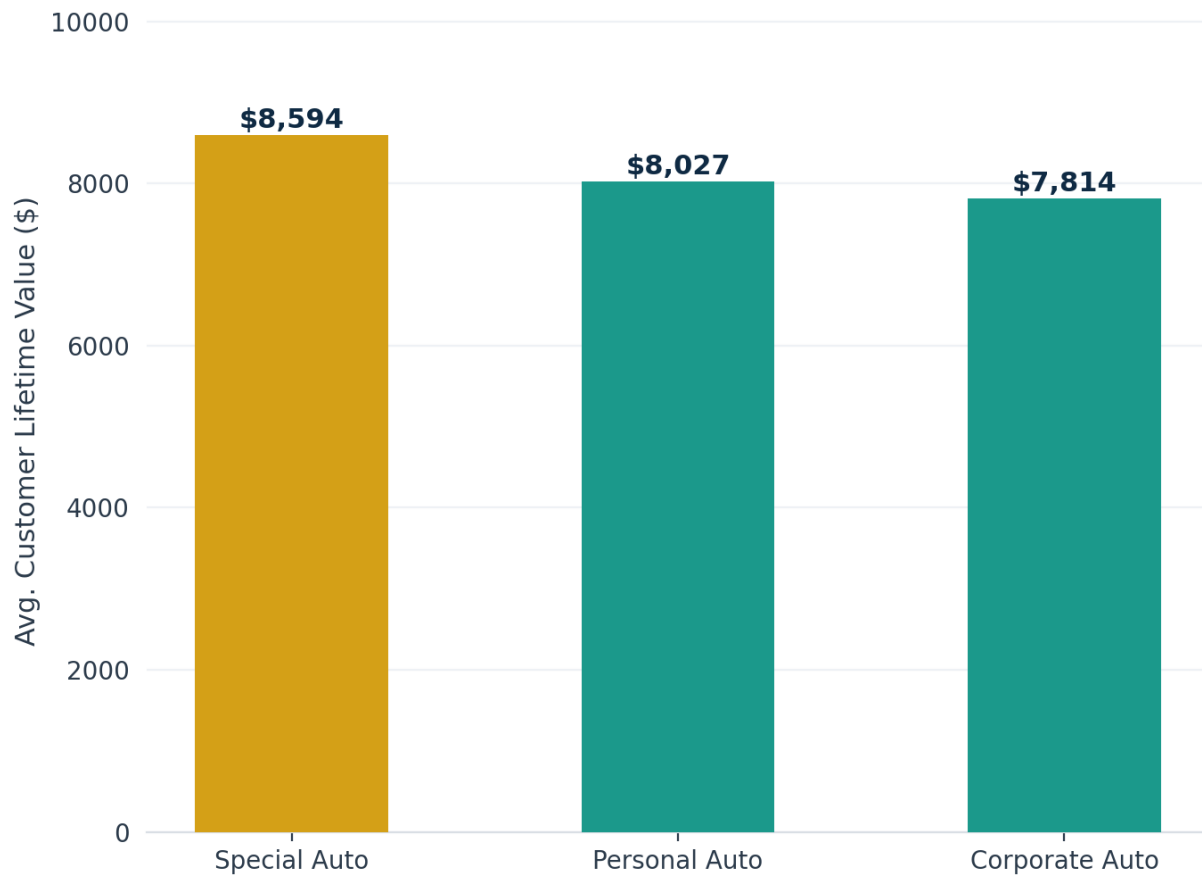
Average CLV was calculated per state, policy type, and vehicle class using GROUP BY, then ranked with window functions (RANK, DENSE_RANK, NTILE) to identify top performers, quartiles, and outlier customers.

Average CLV by State



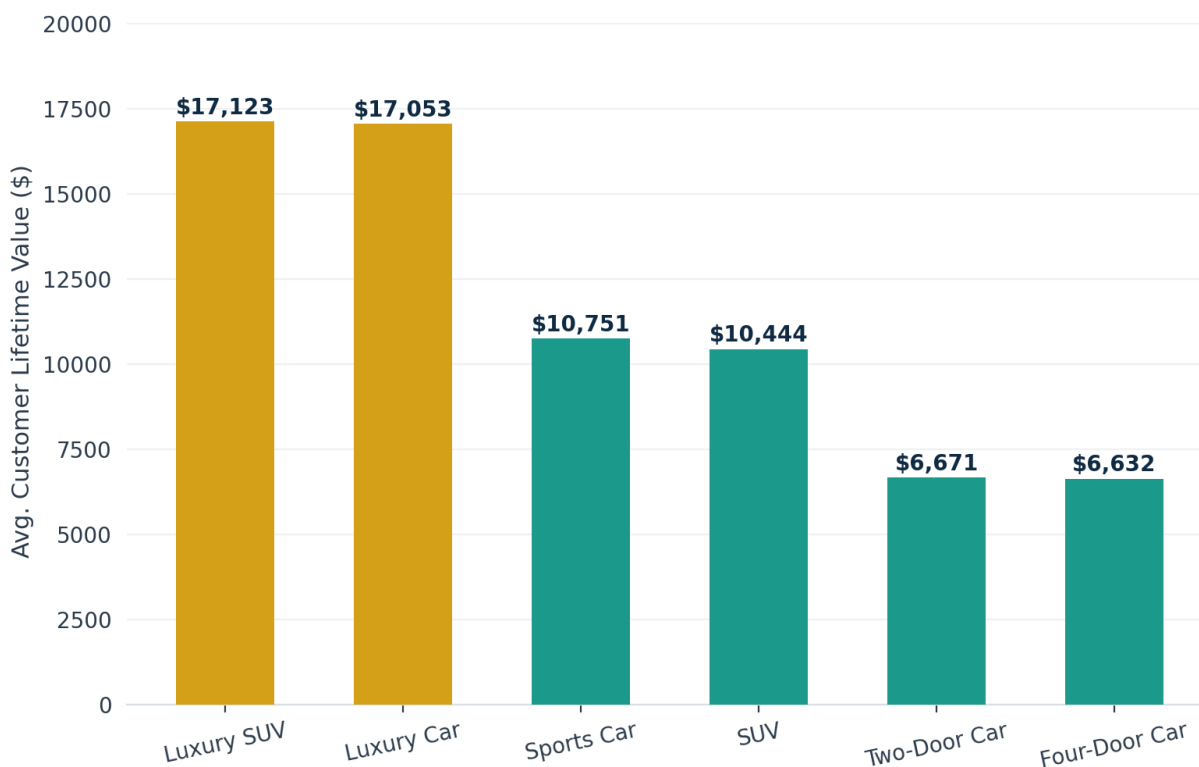
Insight: CLV is remarkably consistent across states — a spread of only about 2.7% between the highest (Oregon, \$8,077.90) and lowest (Arizona, \$7,861.34) average CLV. This suggests customer value is driven far more by policy and vehicle characteristics than by geography.

Average CLV by Policy Type



Insight: Special Auto policyholders carry the highest average CLV (\$8,594.25) despite being the smallest policy segment (378 customers, 4.1% of the base) — a small, high-value niche worth protecting and expanding, rather than the higher-volume Personal Auto segment (\$8,027.36).

Average CLV by Vehicle Class



Insight: Luxury SUV and Luxury Car owners carry roughly 2.5x the average CLV of Four-Door and Two-Door Car owners (\$17,123 / \$17,053 vs. \$6,632 / \$6,671) — vehicle class is one of the strongest CLV predictors in the dataset, even stronger than state or policy type.

CLV Quartile Profile (NTILE(4))

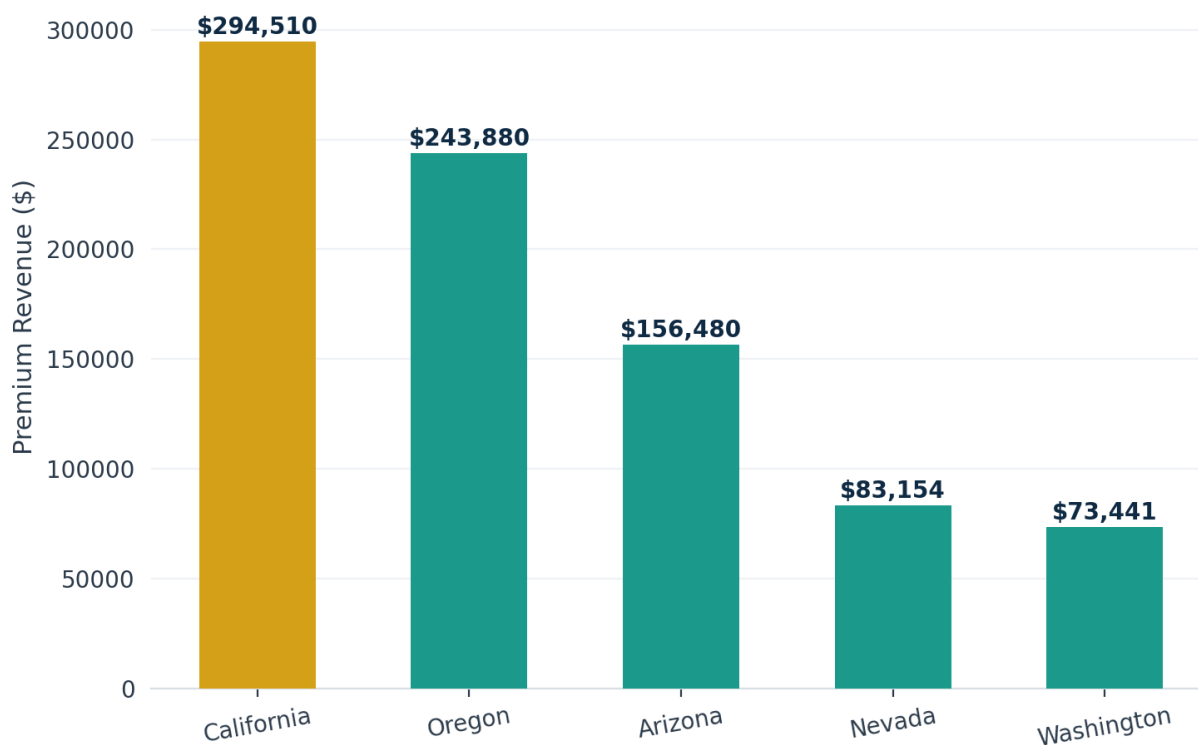
Quartile	Customers	Avg. CLV	Avg. Premium	Avg. Claim	Avg. Income	Avg. Profit
Q1 (Top 25%)	2,284	\$16,722.74	\$116.82	\$547.29	\$38,333.54	\$16,175.45
Q2	2,283	\$7,435.61	\$94.71	\$439.01	\$38,179.59	\$6,996.61
Q3	2,283	\$4,952.03	\$84.90	\$390.26	\$37,997.87	\$4,561.77
Q4 (Bottom 25%)	2,284	\$2,907.79	\$76.45	\$359.78	\$36,118.89	\$2,548.01

Insight: The top CLV quartile is worth 5.7x the bottom quartile in average profit (\$16,175 vs. \$2,548) — yet average income barely differs between them (\$38,334 vs. \$36,119). CLV in this book is driven by policy structure and vehicle choice, not income, meaning income alone is a poor targeting variable for high-value acquisition.

05 Revenue & Premium Analysis

Monthly premium revenue was aggregated by state, policy type, and demographic cuts to identify where premium income concentrates.

Total Premium Revenue by State



Insight: California alone generates **\$294,510** in monthly premium revenue — 34.6% of the company total — consistent with its 34.5% share of the customer base. Revenue concentration mirrors customer concentration almost exactly, meaning growth here comes from volume rather than any regional pricing premium.

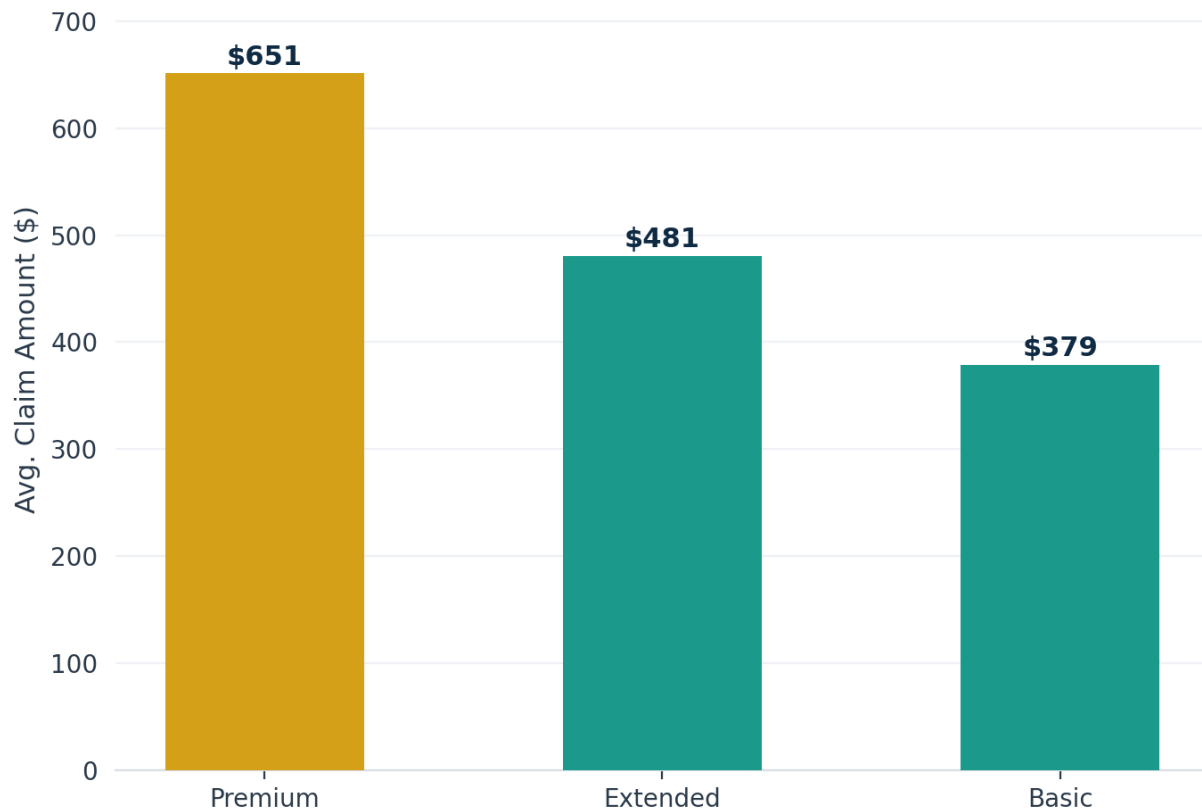
Cut	Highest	Value
Avg. Premium by Marital Status	Single	\$93.88/mo
Avg. Premium by Vehicle Size	Medsize	\$93.53/mo
Total Premium by Education	Bachelor's Degree	\$254,178
Avg. Income by Education	Master's Degree	\$44,768.19
Avg. CLV by Employment Status	Employed	\$8,219.12

Insight: Premium amounts barely vary by marital status or vehicle size (all within a \$2–3 band) — pricing in this book appears driven almost entirely by coverage level and vehicle class rather than these demographic factors, which is worth validating against the actual rating model.

06 Claims Analysis

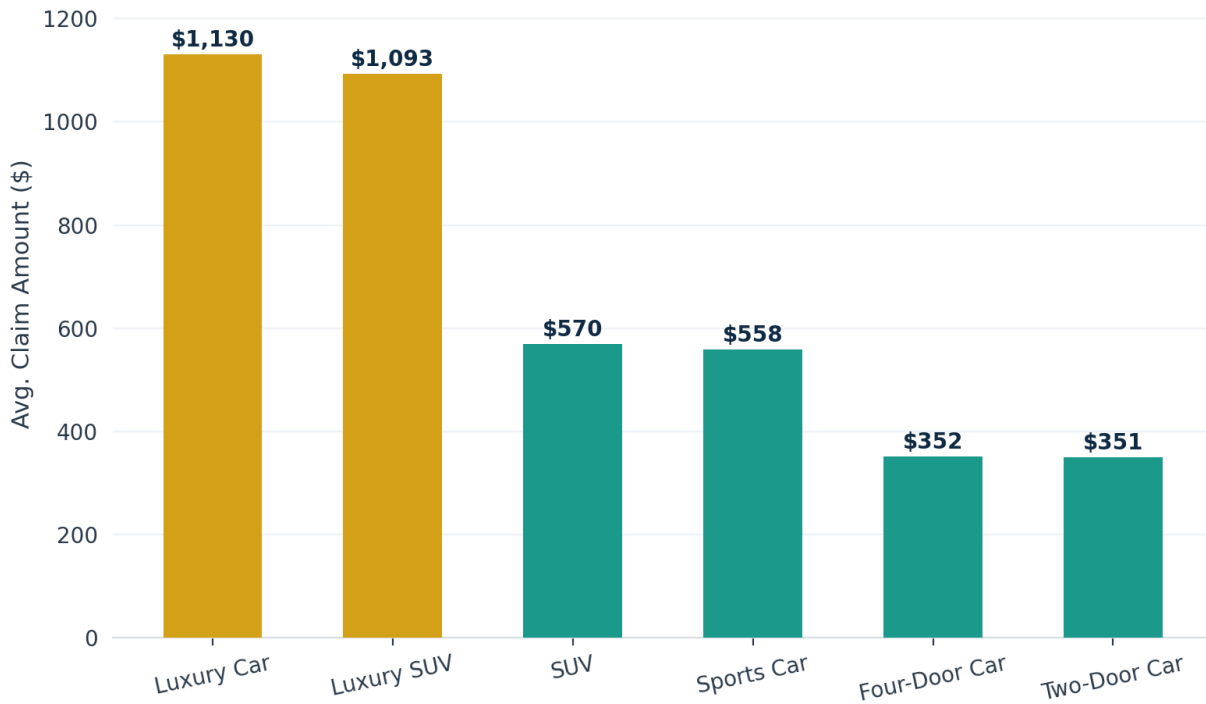
Average claim amount was compared across coverage tier and vehicle class to identify where claims risk concentrates.

Average Claim Amount by Coverage



Insight: Premium coverage customers file claims 72% larger than Basic coverage customers on average (\$651.39 vs. \$379.04) — expected, since Premium coverage insures against more scenarios, but worth confirming premium pricing fully offsets this gap (see Section 8's profitability findings).

Average Claim Amount by Vehicle Class

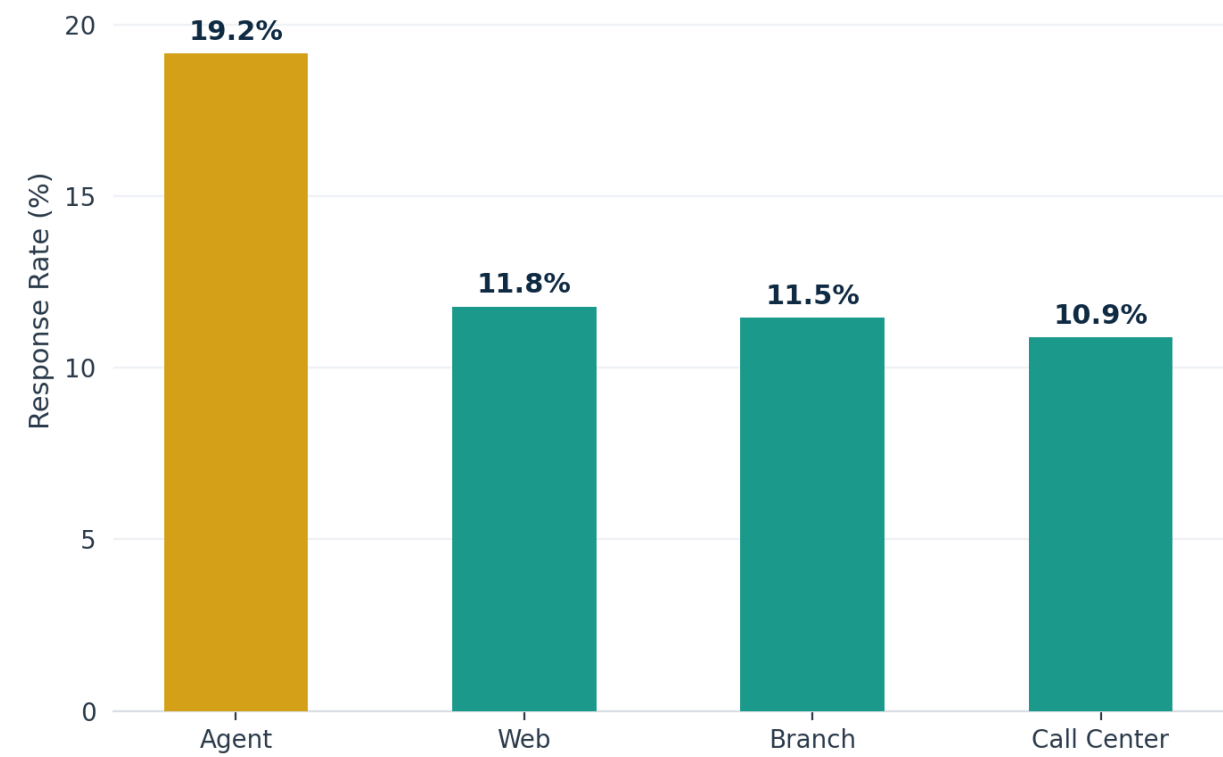


Key Finding: Luxury Car and Luxury SUV owners file claims averaging **\$1,130** and **\$1,093** — roughly 3.2x the average claim for a Four-Door Car (\$351.76). Combined with their high CLV (Section 4), luxury vehicle owners are simultaneously the most valuable and the highest-claims segment.

07 Marketing Response Analysis

Campaign response was analyzed by sales channel, policy type, and renewal offer type to identify which acquisition and retention channels perform best.

Campaign Response Rate by Sales Channel



Key Finding: The Agent channel converts at **19.15%** — nearly double the Web (11.77%), Branch (11.45%), and Call Center (10.88%) channels. Agent-led outreach is the single strongest marketing lever identified in this dataset.

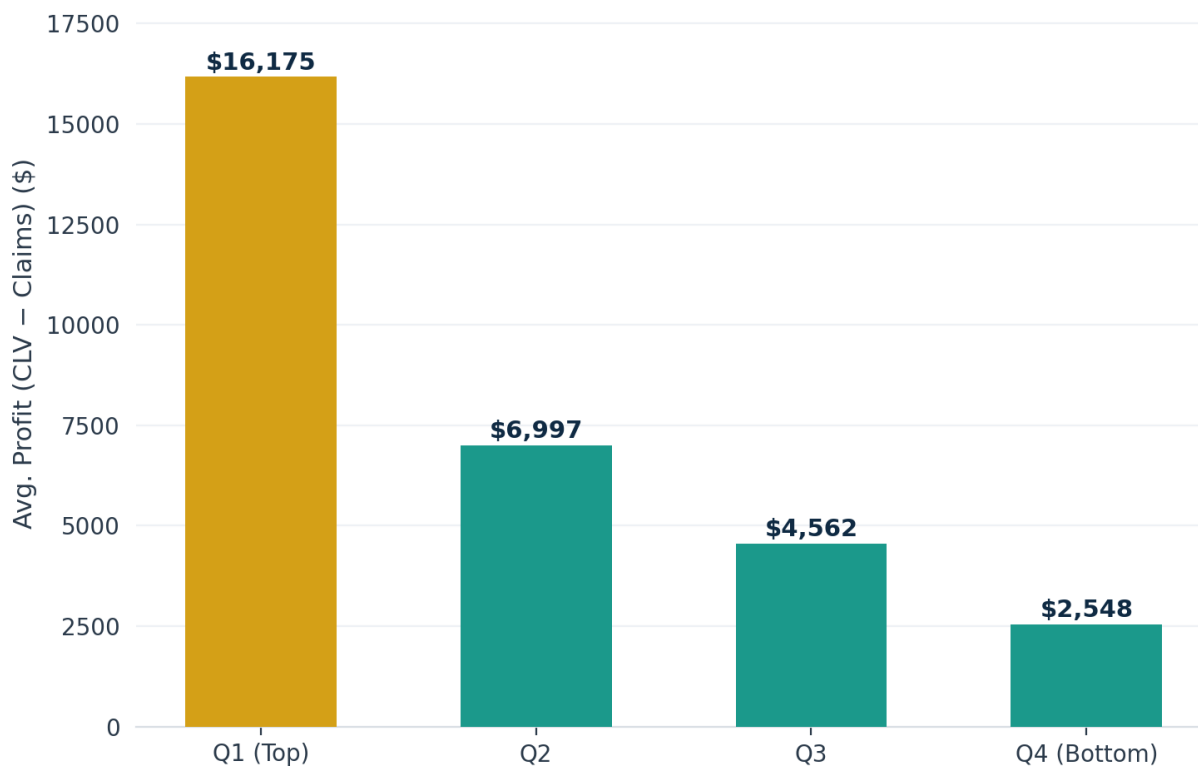
Cut	Finding
Response Rate by Policy Type	Special Auto highest (16.40%), Personal Auto lowest (14.11%)
Response Rate by Renewal Offer	Offer2 highest (23.38%); Offer4 converted 0% of recipients
Avg. CLV — Responders vs. Non-Responders	Non-responders slightly higher CLV (\$8,030 vs. \$7,855)
Avg. Income — Responders vs. Non-Responders	Responders earn marginally more (\$38,544 vs. \$37,509)

Insight: Responders do not have meaningfully higher CLV or income than non-responders — meaning the current campaign is not preferentially reaching high-value customers. This reinforces the opportunity flagged in Section 9: high-CLV non-responders are a distinct, underserved segment, not simply an extension of who already responds.

08 Risk & Profitability Segments

Composite SQL filters were built to flag customers whose premium does not match their underlying value or risk — both under-insured (revenue leakage) and over-insured (retention/pricing risk) segments.

Average Profit per Customer by CLV Quartile



Sample SQL — Underinsured Customer Filter:

```
SELECT customer, customer_lifetime_value, monthly_premium_auto
FROM customer_insurance
WHERE customer_lifetime_value > 40000
  AND monthly_premium_auto < 120
ORDER BY customer_lifetime_value DESC;
```

Risk Segment	Definition	Customers Flagged
Underinsured	CLV > \$40,000 AND Premium < \$120/mo	13
Overinsured	CLV < \$10,000 AND Premium > \$100/mo	2,003
Most Profitable Coverage	Premium tier	\$10,244 avg. profit/customer

Key Finding: 2,003 customers — nearly 22% of the entire book — are overinsured: paying above-average premiums (\$100+/mo) while carrying below-average CLV (under \$10,000). This is the single largest structural finding in the dataset and represents a major retention and re-pricing risk if these customers ever shop their policy.

Insight: Despite filing the largest claims (Section 6), Premium coverage remains the most profitable tier per customer (\$10,244 avg. profit vs. \$6,812 for Basic) — the higher premium charged more than compensates for the higher claims cost.

09 High-Value Target List

A composite filter identifies the highest-priority marketing target: customers with CLV above \$40,000 who have not yet responded to the current campaign — a strong-fit, low-effort-to-convert segment.

Action Required: 54 customers meet both criteria — CLV over \$40,000 and no response to the current campaign. This is a small, addressable list that should be prioritized for a targeted, high-touch (ideally Agent-led, per Section 7) follow-up campaign rather than blanket outreach.

Top 10 Customers by Customer Lifetime Value

Customer ID	State	Customer Lifetime Value	Monthly Premium
FQ61281	Oregon	\$83,325.38	\$231
YC54142	Washington	\$74,228.52	\$242
BP23267	California	\$73,225.96	\$202
KH55886	Oregon	\$67,907.27	\$192
SK66747	Washington	\$66,025.75	\$188
FB95288	California	\$64,618.76	\$217
AZ84403	Oregon	\$61,850.19	\$238
US30122	California	\$61,134.68	\$198
JT47995	Arizona	\$60,556.19	\$204
EN65835	Arizona	\$58,753.88	\$185

Sample SQL — Top-1% Customer Identification (NTILE(100)):

```
SELECT customer, customer_lifetime_value FROM (
  SELECT customer, customer_lifetime_value,
    NTILE(100) OVER (ORDER BY customer_lifetime_value DESC) AS percentile
  FROM customer_insurance
) AS ranked_customers WHERE percentile = 1 ORDER BY customer_lifetime_value DESC;
```

10 Consolidated Business Insights

- The portfolio is worth **\$73.1M** in total CLV against **\$851K** in monthly premium and **\$3.96M** in claims — a healthy overall margin, but concentrated unevenly across segments.
- **2,003 customers (22%)** are overinsured — paying above-average premium against below-average CLV — the largest structural risk identified in this analysis.
- Vehicle class is the strongest CLV predictor: Luxury SUV/Car owners carry **2.5x** the CLV of standard sedan owners, but also file **3.2x** larger claims.
- Special Auto is the smallest policy segment (4.1% of customers) but carries the **highest average CLV** (\$8,594) — a high-value niche worth protecting.
- The **Agent channel converts at nearly 2x** every other sales channel (19.15% vs. 10.9–11.8%) — the clearest marketing lever in the dataset.
- Premium coverage customers file the largest claims per tier, yet remain the **most profitable tier overall** (\$10,244 avg. profit/customer) — pricing already compensates for risk.
- CLV is nearly flat across states (2.7% spread) and across income levels (Q1 vs Q4 income differs by only 5.7%) — geography and income are weak targeting variables; policy and vehicle class are strong ones.
- **54 high-CLV customers** (>\$40,000) have not responded to the current campaign — a small, high-value, immediately actionable target list.
- Only **13 customers** are underinsured (high CLV, low premium) — a minor revenue-leakage risk compared to the overinsured segment.
- Overall campaign response rate is a modest **14.32%**, and responders are not meaningfully higher-value than non-responders — the current campaign is not yet preferentially reaching high-CLV customers.

11 Recommendations

Priority	Recommendation	Target Area
High	Launch a retention review for the 2,003 overinsured customers — reassess pricing or coverage fit before they shop competitors.	Retention / Pricing
High	Run a targeted, Agent-led campaign for the 54 high-CLV non-responders identified in Section 9.	Marketing / High-Value Targeting
High	Shift incremental marketing spend toward the Agent channel given its ~2x conversion rate over Web, Branch, and Call Center.	Channel Strategy
Medium	Protect and grow the Special Auto segment — smallest volume, highest CLV per customer.	Segment Strategy
Medium	Review claims-adjudication and risk controls for Luxury Car/SUV owners given their outsized claim severity.	Claims / Underwriting
Medium	Validate that Premium-tier pricing continues to outpace claims growth as that segment scales.	Pricing Strategy
Medium	Re-evaluate renewal Offer4 (0% response rate) — likely candidate for retirement or redesign.	Campaign Design
Low	Build a recurring CLV/claims/response dashboard (state, policy, coverage, channel) to monitor these KPIs monthly.	Analytics Ops

12 Methodology & Conclusion

Methodology

- Data preparation: the raw dataset was loaded into a PostgreSQL table (customer_insurance) with all 24 source columns; duplicate and NULL checks confirmed a clean dataset.
- Exploratory queries: basic SELECT/WHERE/ORDER BY queries profiled states, policy types, coverage, and customer demographics.
- Aggregation: COUNT, AVG, SUM, MIN/MAX with GROUP BY quantified customer distribution, CLV, premium, and claims by segment.
- Conditional logic: CASE WHEN/FILTER statements built response-rate calculations and null-handling logic.
- Subqueries & CTEs: identified above-average earners, above-average state CLV, and customers exceeding their state's average claim.
- Window functions: ROW_NUMBER, RANK, DENSE_RANK, NTILE, LAG/LEAD ranked customers within states, built CLV/income/premium quartiles and deciles, and compared sequential CLV benchmarks.
- Business synthesis: results from all query stages were consolidated into the insights (Section 10) and recommendations (Section 11) presented in this report.

Conclusion

This SQL-based analysis of the IBM Watson Marketing Customer Value dataset demonstrates how layered querying — from simple filters through aggregation, subqueries, CTEs, and window functions — can convert raw insurance records into an evidence-based retention and marketing strategy. The data consistently points to overinsured customers, vehicle-class risk concentration, and underused Agent-channel conversion as the primary levers for improving both profitability and campaign performance. Acting on the prioritized recommendations in Section 11, and operationalizing the KPI queries built in this project, gives the business a repeatable, data-driven framework for customer value management going forward.

Sources: WA_Fn-UseC_-Marketing-Customer-Value-Analysis.csv (raw data, IBM Watson Analytics sample dataset), CLV.sql (PostgreSQL query script).

Appendix — Full-Size Charts

All visualizations from this report, provided at full size for reference or reuse.

Average CLV by State

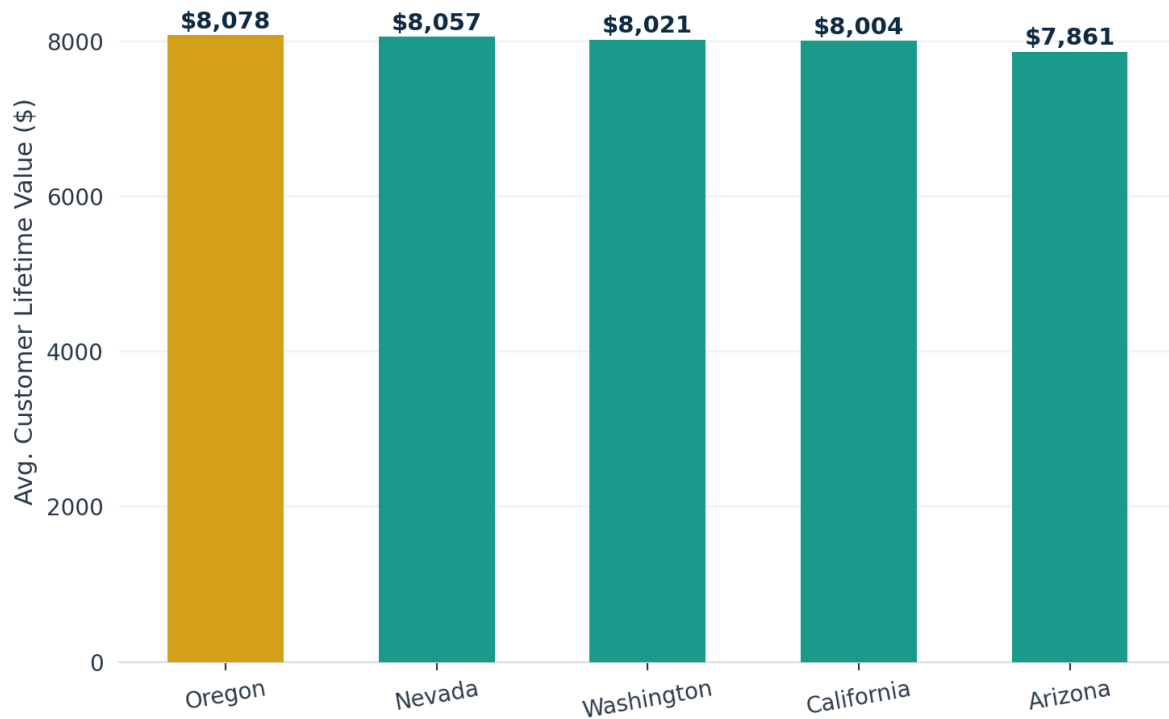


Figure 1. Average CLV by State

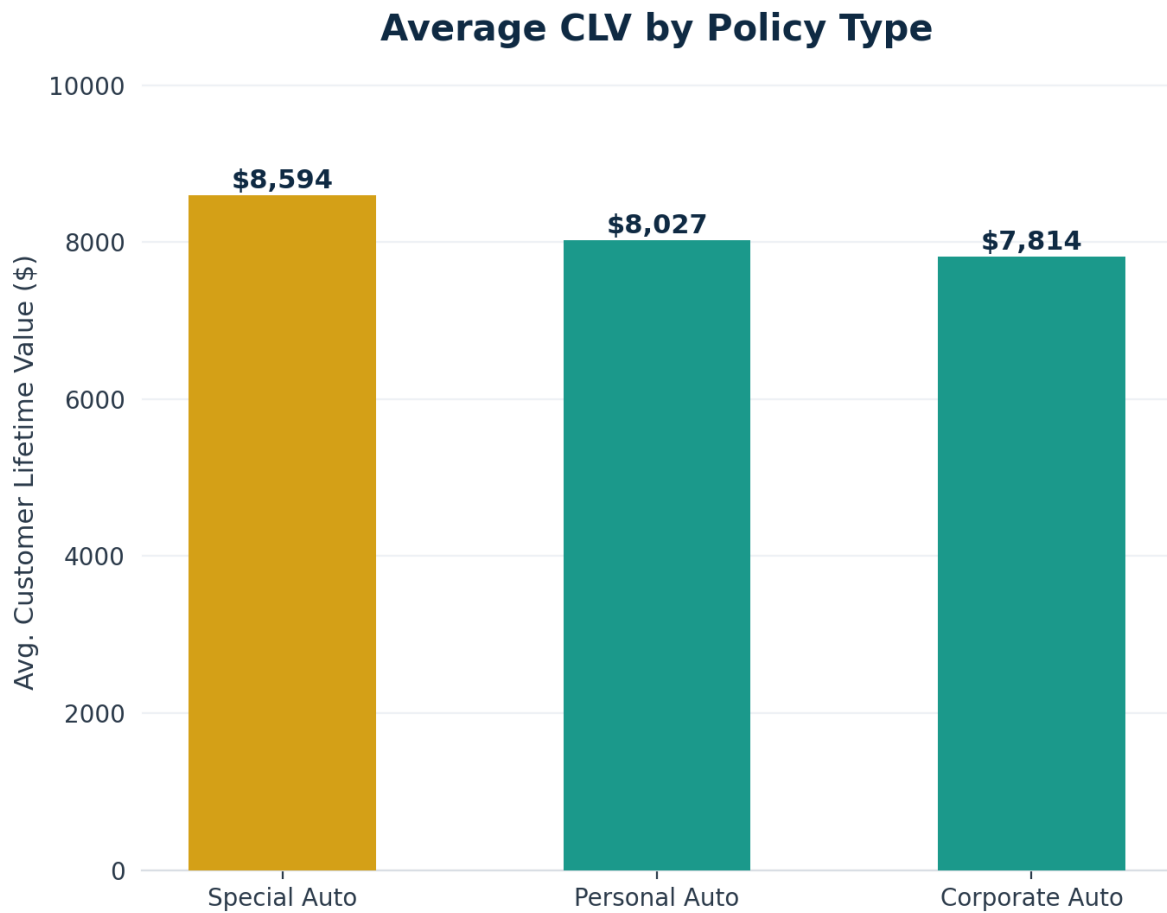


Figure 2. Average CLV by Policy Type

Total Premium Revenue by State

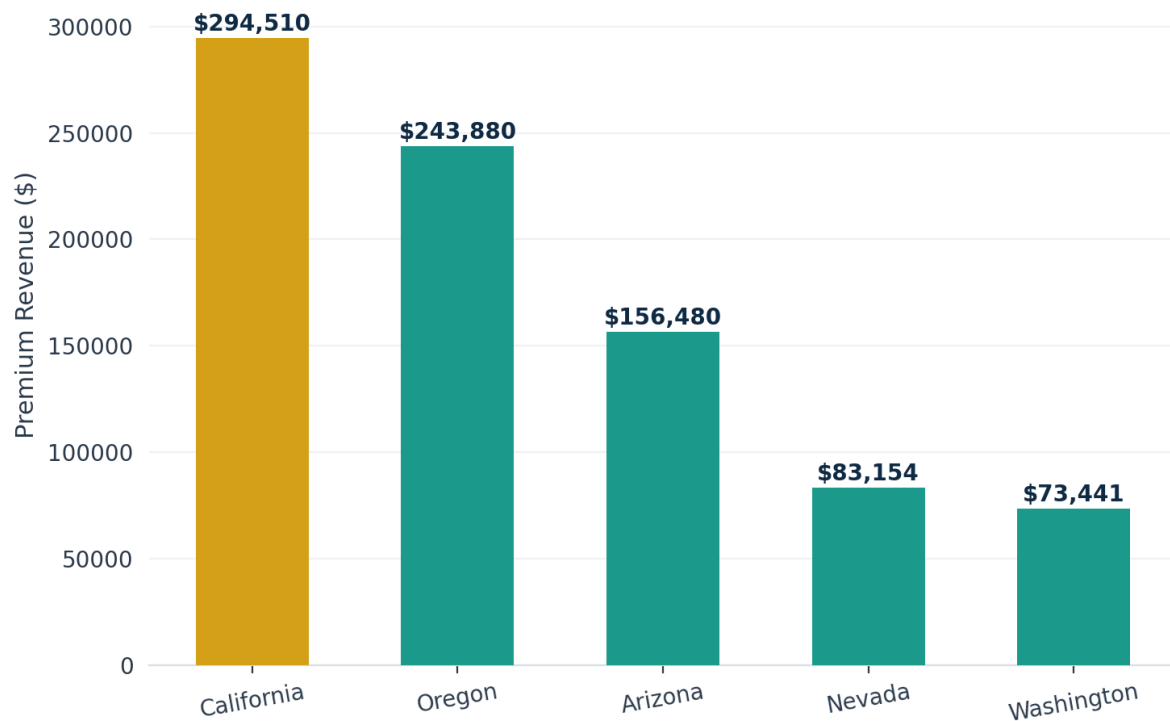


Figure 3. Total Premium Revenue by State

Customers by Coverage Level

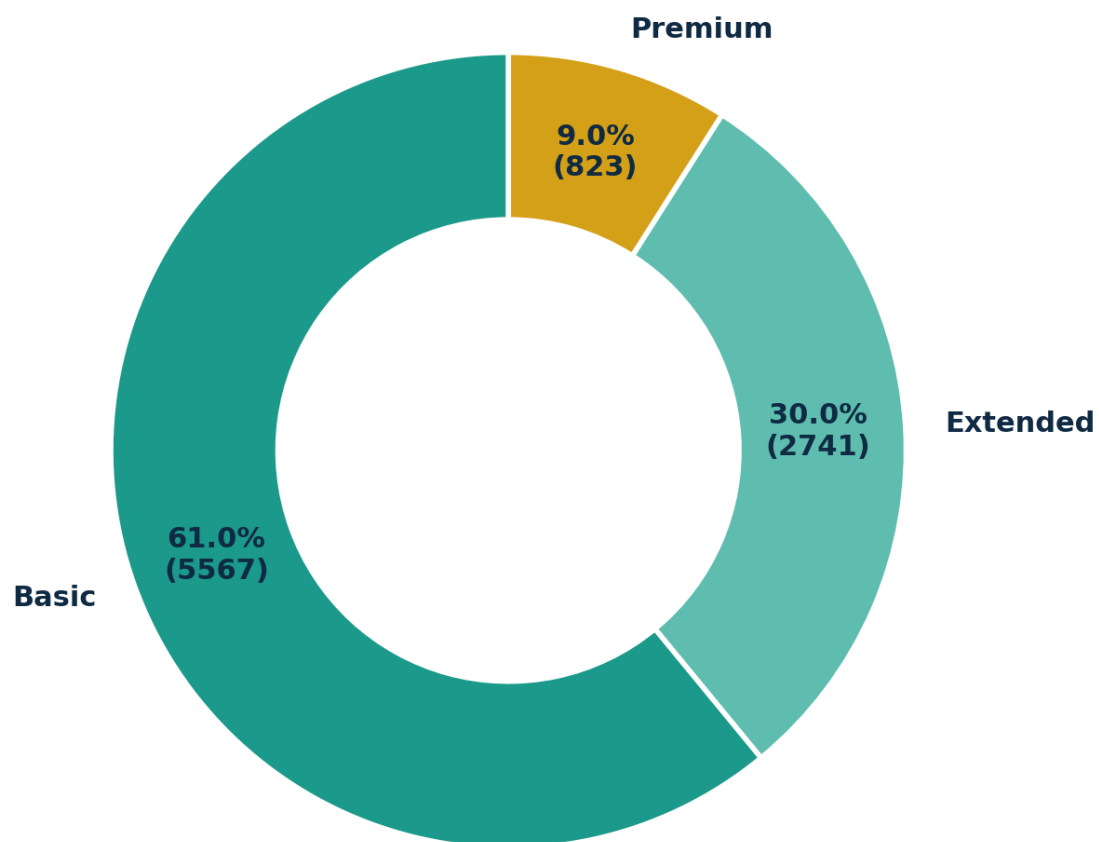


Figure 4. Customers by Coverage Level

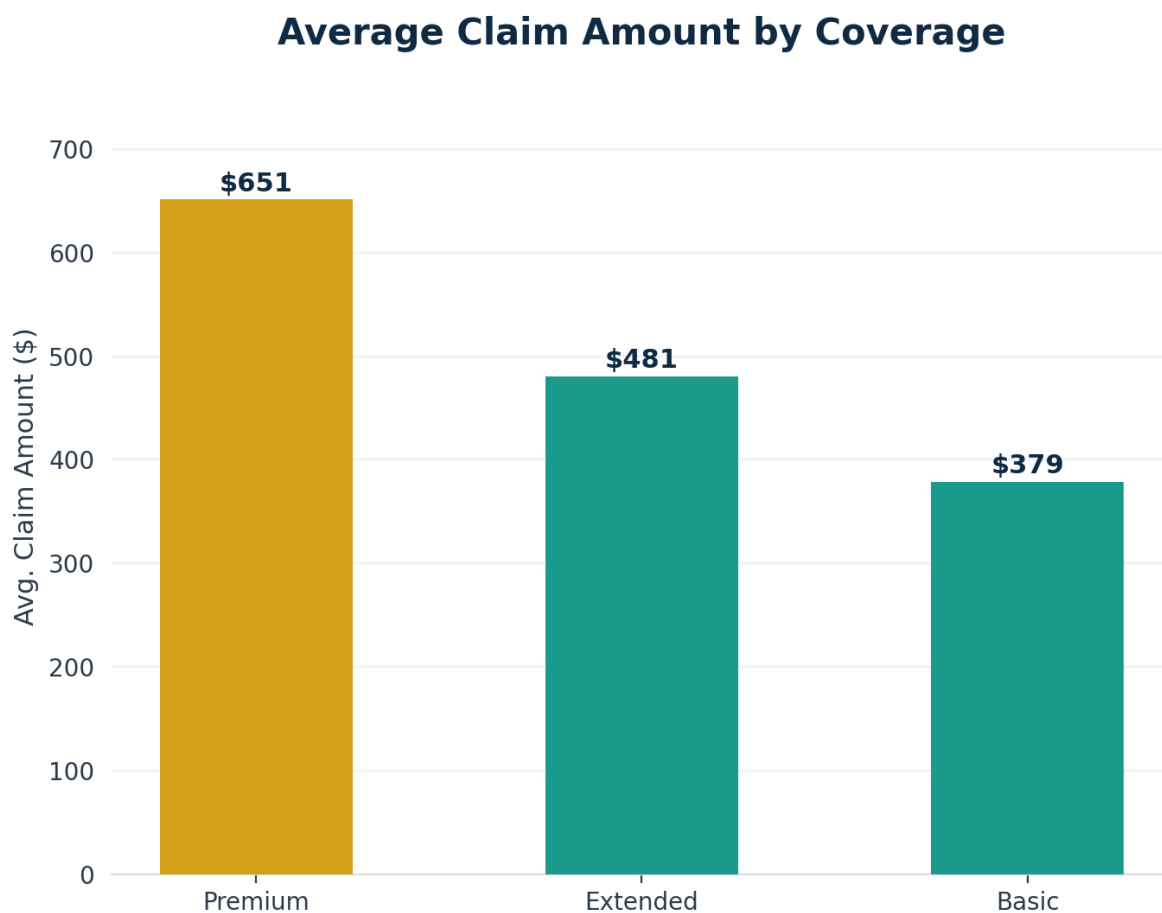


Figure 5. Average Claim Amount by Coverage

Average Claim Amount by Vehicle Class

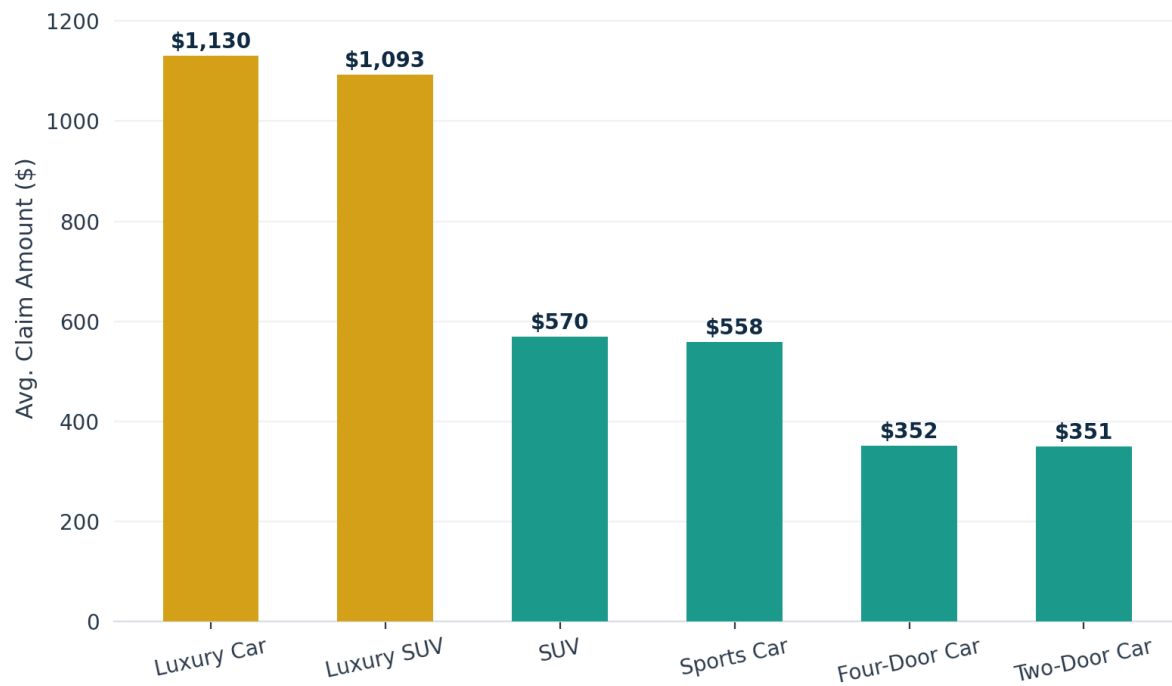


Figure 6. Average Claim Amount by Vehicle Class

Campaign Response Rate by Sales Channel

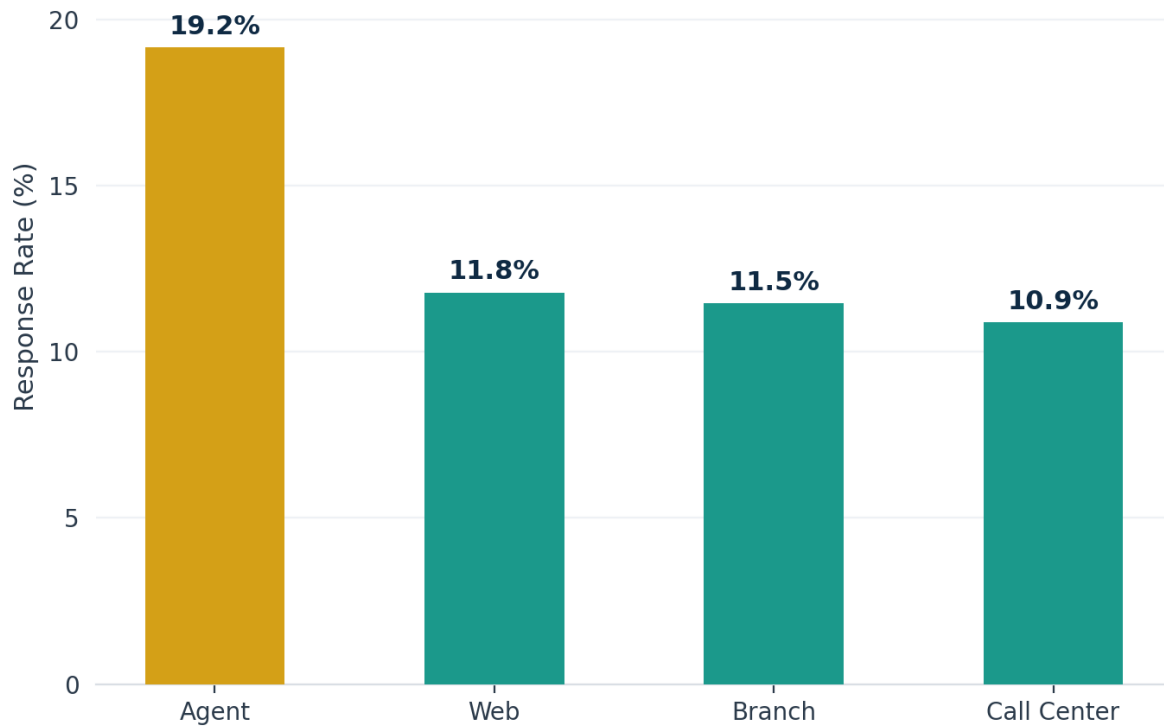


Figure 7. Campaign Response Rate by Sales Channel

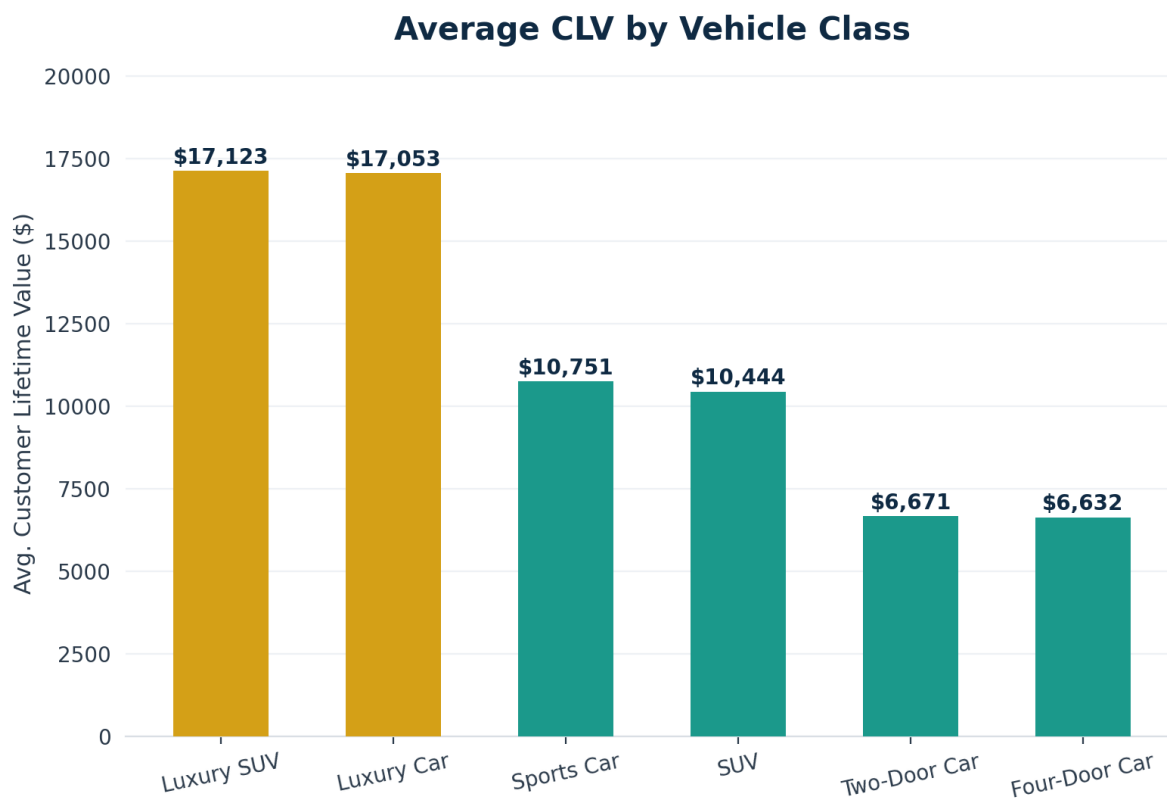


Figure 8. Average CLV by Vehicle Class

Average Profit per Customer by CLV Quartile

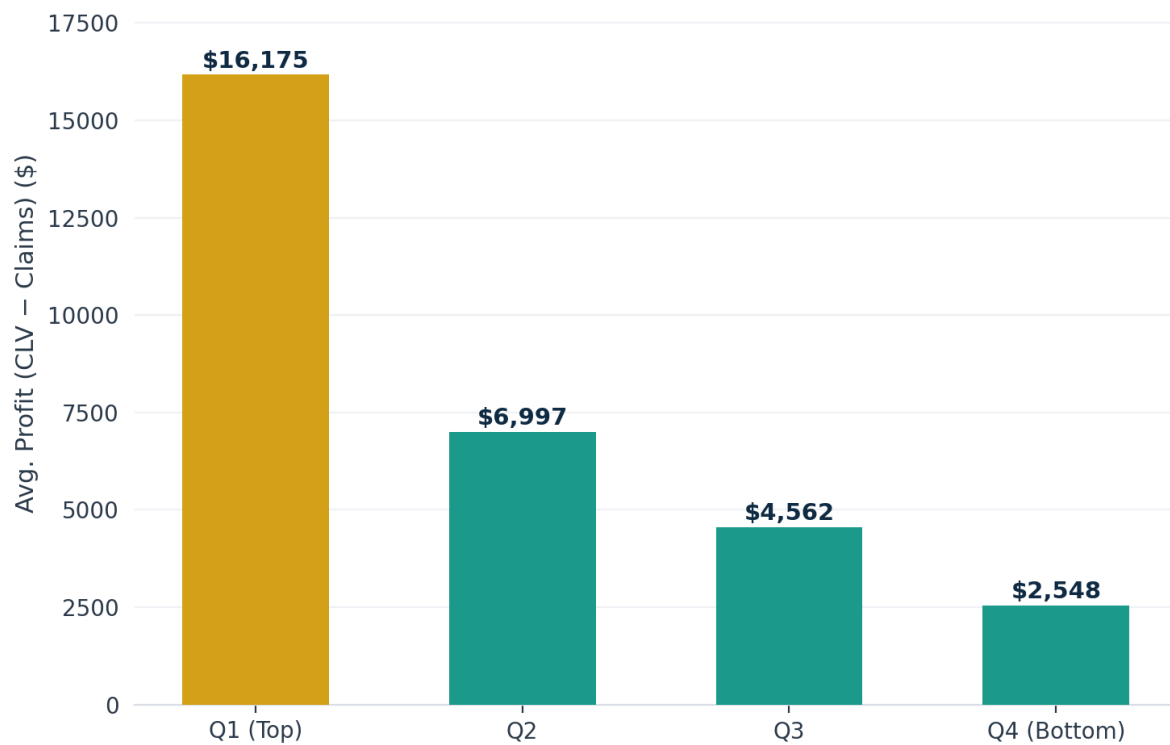


Figure 9. Average Profit per Customer by CLV Quartile